

Shashank Chakilam

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Design Verification Engineer | UVM/SVA | AXI | PCIe | DDR | UPF | CDC/RDC | Python

Professional Summary

Design Verification engineer with 2 years of UVM/SystemVerilog experience. Strong in SVA, constraint-random, coverage closure, and regression automation on Linux. Power-aware verification (UPF), CDC/RDC, and GLS familiarity. Wrote testplans, covergroups, and checkers; closed high functional/code coverage on complex IP.

Skills

Methodology: UVM (agents, sequences, scoreboards), SVA, constraint-random, functional/code coverage, testplan authoring, regressions, seeding

Power/Signoff: UPF (isolation/retention), CDC/RDC, lint, GLS

IPs/Interfaces: AXI4/AXI4-Lite (project), APB/AHB, I2C, SPI; familiarity with PCIe/CXL, DDR/LPDDR, NoC

EDA Tools: Synopsys VCS, Verdi; Cadence SimVision/DVE; PrimeTime; Design Compiler

Programming: Python, SystemVerilog, C, Perl; Git; Jira; Linux/Unix

Work Experience

Netcracker Technology, ASIC Verification Engineer

Mar 2021 – Dec 2022

- Verified low-power features for a PCH-class SoC: power-gating/clock-gating, sleep/reset/wake sequences, and PLL bring-up; authored SVAs for isolation/retention and reset ordering (UPF-aware checks).
- Built Python tooling to triage failing consumer clocks and aggregate regression results; cut manual log review by ~90%.
- Drove coverage closure for APB-attached IP: constraint-random sequences, protocol checks, and cross coverage; documented last-mile bins and fixes.
- Ran nightly regressions on VCS/Verdi; filed/triaged bugs with designers; tracked closure to sign-off criteria.

Projects

Motion-Estimation Engine (14 nm)

- UVM testplan; reusable agent for 16×16 block matching; 40+ SVAs for handshakes, bounds, and pipeline stalls.
- Closed 95% functional and 100% code coverage in VCS; root-caused 17 corner bugs with waveform + assertion hits.

AXI4-Lite Interconnect Verification (Mini-SoC)

- Built UVM environment for an AXI4-Lite slave behind a simple interconnect; agents, sequences, scoreboard, protocol checkers.
- Constraint-random traffic across address, byte-enables, and back-to-back operations; cross coverage on `aw_prot` × back-to-back writes × resets.
- SVAs for valid/ready handshakes, ordering, and exclusivity; reached 95% functional and 100% code coverage; 500+ nightly tests via Python CI.

MDIO Interface Verification

- Developed end-to-end IEEE 802.3 Clause22 testplan with protocol checkers, error-injection, Covergroups for opcodes.
- Integrated agent into SystemVerilog environment; reached 90% functional coverage.

Fixed-Point Arithmetic Core

- From-scratch UVM env (generator, sequencer, driver, monitor, scoreboard); coverage-driven closure of rounding/saturation corners.
- Employed a coverage-driven methodology, using functional coverage and SVA to validate complex rounding and saturation logic, with 90% functional and 100% code coverage.

Education

Master of Science in Electrical and Computer Engineering

Jan 2023 – Aug 2025

San Francisco State University

GPA: 3.7

Coursework: Advanced ASIC Verification, VLSI Physical Design, Computer Architecture (Arm and RISC-V), Digital System Testing, Embedded Systems Programming, Low-Power Design Techniques, Static Timing Analysis

Bachelor of Science in Electronics and Communication Engineering

Aug 2017 – May 2021

JNTUH, KMIT

GPA: 3.5

F-1 OPT (STEM). Open to Immediate relocation.